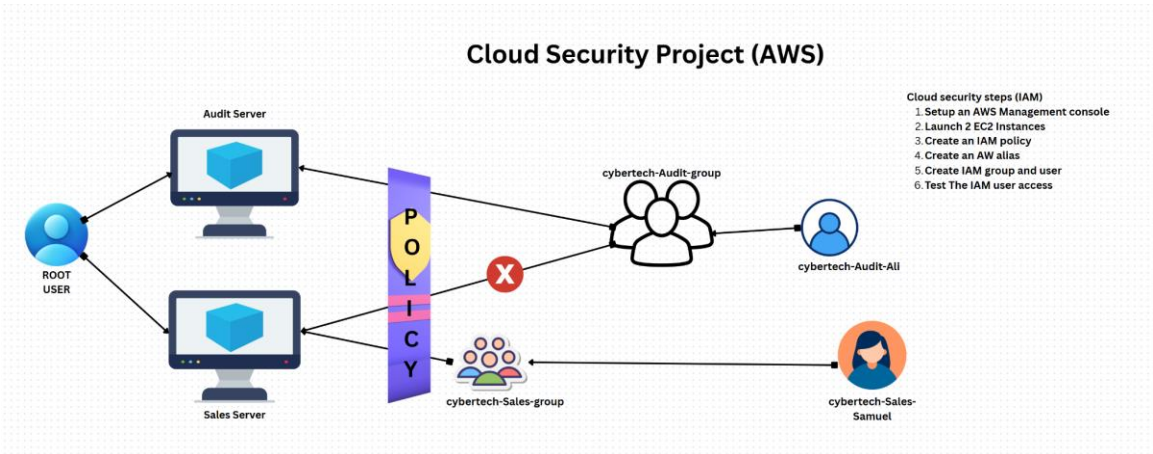


AWS IAM Cloud Security Project

1. Project Overview

I completed this project on cloud security controls in Amazon Web Services (AWS), focusing on Identity and Access Management (IAM). The goal was to create a least-privilege policy, attach it to a user group, and verify that the policy correctly restricts actions on two Amazon EC2 instances (audit and sales).



2. Tools & Concepts

- AWS IAM – users, groups, policies, account alias
- Amazon EC2 – instance tagging and lifecycle actions
- JSON policy syntax – Effect, Action, Resource
- Principle of least privilege and policy testing

3. Tagging Strategy

I applied a descriptive tag to each EC2 instance:			
Instance	Tag Key		Tag Value
audit		Environment	Audit
sales	Environment Sales		

Instances (1/2) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

Instance state: [running](#) [Clear filters](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☐	cybertech-aud...	i-03e590b416cd54355	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-61
☒	cybertech-sale...	i-0c4adcc4d506a3321	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-16-17

4. Creating the IAM Policy

I authored the following JSON policy to block instance stop/start actions on the audit server but allow those actions on the sales server:

Permissions defined in this policy [Info](#) [Copy](#) [Edit](#) [Summary](#) [JSON](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "Audit"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
```

5. Account Alias

I set a memorable account alias to replace the default numeric URL, making sign-in easier for team members.



AWS Account

Account ID

 288645738432

Account Alias

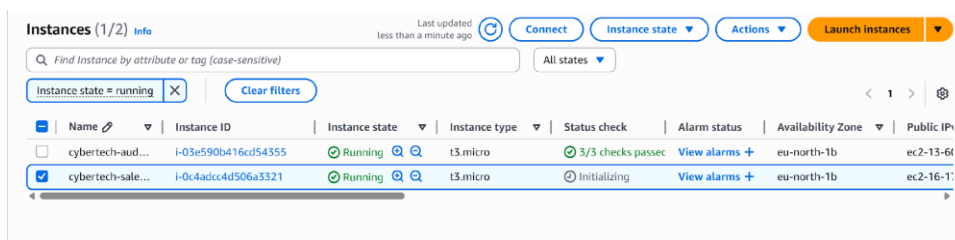
cybertechusers [Edit](#) | [Delete](#)

Sign-in URL for IAM users in this account

 <https://cybertechusers.signin.aws.amazon.com/console>

6. IAM Users & Groups

1. Created an IAM user group called Developers.
2. Attached the **CybertechAuditEnvPolicy** policy to the group.
3. Added individual IAM users who require controlled EC2 access.

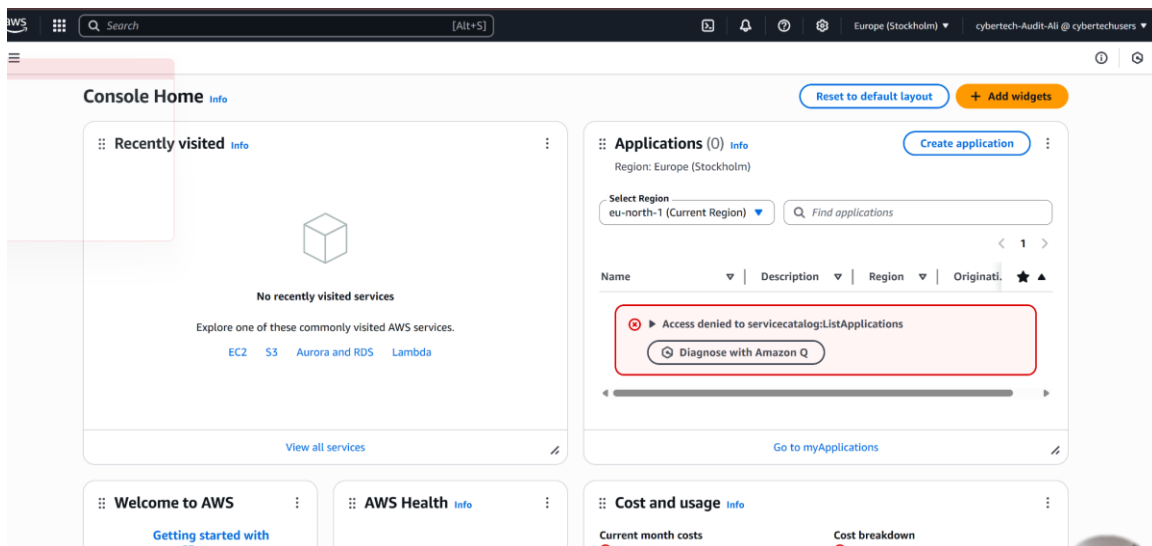


The screenshot shows the AWS Management Console 'Instances' page. It displays a table of EC2 instances. The first instance, 'cybertech-aud...', is in a 'Running' state. The second instance, 'cybertech-sale...', is in an 'Initializing' state. The table includes columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IP.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
cybertech-aud...	i-03e590b416cd54355	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-64...
cybertech-sale...	i-0c4adcc4d506a3321	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-16-1...

7. Logging in as an IAM User

- IAM users can sign in through:
- AWS Management Console (using the new alias URL)
 - AWS CLI via programmatic keys



8. Testing the Policy

Test	Action	Expected	Result	Actual	Result
Stop	audit instance	Denied	Access denied	error	displayed
Stop	sales instance	Allowed	Instance stopped	successfully	
Start	audit instance	Denied	Access denied	error	displayed
Start	sales instance	Allowed	Instance started	successfully	

IAM Dashboard Info

Security recommendations 0

⊗ Access denied

You don't have permission to `iam:GetAccountSummary`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali

Action: iam:GetAccountSummary

Context: no identity-based policy allows the action

 Diagnose with Amazon Q

⊗ Access denied

You don't have permission to `iam:ListMFADevices`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali

Action: iam:ListMFADevices

Context: no identity-based policy allows the action

⊗ Failed to stop the instance i-0b9626f2835a5f977

You are not authorized to perform this operation. User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:288645738432:instance/i-0b9626f2835a5f977 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: sAveffO1FXITMRJ52xofnO4bpSGLz7IFqT-w1DqTxLoat_IV4T5r87OVMMHxzfSSmG4VXfVa2cdWhfQ1uQO0shFk8kUgfZhoC9Tn_lhCnKyVswG-blgkFFw9OLb4-2CT9TP841I8ni9O6EaSTAt0ztgtsh2t1VXnMzhUuPZeqMg3Ajek6wr21590c0le6YNjRMmccGw8Gd_xH9dlfdp-4ilwlF0H1uwTpAt1QHqqr8FbW3kjOue7UMNuHGzJzgCiiA30Q5pqHwW1_ALxzQAulhMpW9i6xdEiwRRLOfQG_W-dDTm6Z7YNPth09weBuulXatLbc13Mz5-cIFyB_0PBgVz8xKMy4kuncL_zDmT3Nfyv0sszpKwvYxm8Du71JN3r5TmByIKAIW0cA0nzcOFVWIQ9kigwxbT-gocngZgZGILqW2RxlqW8XBC42OEN3iIz-URe-xY2CS1palfmclLRIDvCBZjNWP4rK_0WgaADMkzFgQjk7xmnnNmFi2mgHr5Aw46VvHCp-pqURgYZ_Oa8xBnq7y6luLZ_MxlUb20pZDhB0SynHoNGRmOMRzRy1etyCipN14aBfvWB12hSkMoXobUZawc9Rat-Pk_vPSbim-Opv1riuGTCT-YJlIZ6FSYUFEcxnmX6B3iFHggGT0p6sDazIXMCX3g2yqEqbTNSf-CIWszbvA7Ck5mSyf8nt1po_-8fBDM67ZwyEe21nGxikjEmj_nMMMMQVPqk46tBhdwRYHLxQQA2b-cG58fOMUsv_eHnT1xBHLtkVMB7VxU-_Scg_KyhwQIMrGy4NR4Ho_RumoDyK8aWNLTixh-j1yTKrJk8ctDTf11QzyfSAKRTQp_-j1QBrzXEA47t8IXAwqRD3vWCNE3PL3YfwcYpRHP

 Diagnose with Amazon Q

Instances (1/2) Info

All states

i-0b9626f2835a5f977 (Cybertech-sales-Ali)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)